Multivariate Data Analysis - MESIO UPC

Matrix decompositions

Jamie Arjona

13/02/2025

UNIVERSITAT POLITÈCNICA
DE CATALUNYA
BARCELONATECH

Presentation outline

- Introduction.
- The Singular Value Decomposition.
- Other decompositions.

Matrix decompositions

- **Matrix decompositions** are fundamental tools in multivariate analysis.
- They allow for the **approximation** of high-dimensional data using a small number of dimensions.
- Given a data matrix $\mathbf{X}_{n\times p}$, how can we find an optimal lower-rank approximation $\hat{\mathbf{X}}_{n\times p}$?
- A primary goal of **dimensionality reduction** is to obtain a lower-dimensional representation that facilitates visualization and reveals underlying patterns in the data.

Matrix decompositions

- There are multiple decompositions but we will put our focus on two of them:
- The spectral decomposition.
- The singular value decomposition (SVD).
- The choice of these decompositions is made because many multivariate methods (PCA, CA, MDS, CCA, etc.) are based on them.

The data

	husband age	husband height	wife age	wife height
1	49	1809	43	1590
2	25	1841	28	1560
3	40	1659	30	1620
4	52	1779	57	1540
5	58	1616	52	1420
6	32	1695	27	1660
7	43	1730	52	1610
8	47	1740	43	1580
9	31	1685	23	1610
10	26	1735	25	1590

Age and height of husband and wife for 10 couples

The data

- The data matrix $\mathbf{X}$ is 10×4 and of rank 4.
- We will approximate it with data matrix $\hat{\mathbf{X}}$ that will be as "close" as possible to $\mathbf{X}$.
- The obtained approximation when used as a rank 2 matrix can be used for visualizations.

Two rank approximation

$$\mathbf{X} = \begin{pmatrix} 49 & 1809 & 43 & 1590 \\ 25 & 1841 & 28 & 1560 \\ 40 & 1659 & 30 & 1620 \\ 52 & 1779 & 57 & 1540 \\ 58 & 1616 & 52 & 1420 \\ 32 & 1695 & 27 & 1660 \\ 43 & 1730 & 52 & 1610 \\ 47 & 1740 & 43 & 1580 \\ 31 & 1685 & 23 & 1610 \\ 26 & 1735 & 25 & 1590 \end{pmatrix} \qquad \hat{\mathbf{X}} = \begin{pmatrix} 43.94 & 1809.21 & 43.72 & 1589.89 \\ 46.36 & 1838.22 & 48.00 & 1562.03 \\ 35.09 & 1659.32 & 29.27 & 1619.79 \\ 44.07 & 1780.43 & 44.75 & 1538.92 \\ 39.40 & 1618.03 & 39.32 & 1418.55 \\ 35.64 & 1694.60 & 29.49 & 1660.28 \\ 39.24 & 1731.53 & 36.00 & 1608.81 \\ 40.67 & 1740.69 & 38.73 & 1579.51 \\ 36.67 & 1683.96 & 31.92 & 1610.78 \\ 39.90 & 1733.25 & 37.32 & 1591.27 \end{pmatrix}$$

Criterion

- The idea is to **minimize the least squares criterion** when computing the approximation.
- In linear regression, the parameters of the model $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$ are estimated by minimizing $\sum e_i^2$ having $e_i = y_i - (b_0 + b_1 x_i)$
- Now, what we do is to minimize the errors in $\mathbf{E} = \mathbf{X} - \hat{\mathbf{X}}$.
- It turns out that the least squares criterion in this case is:

$$\sum_{i=1}^{n} \sum_{j=1}^{p} e_{ij}^2 = \mathrm{tr}(\mathbf{E}^\top \mathbf{E})$$

- $||\mathbf{E}||_F^2 = \mathrm{tr}(\mathbf{E}^\top \mathbf{E})$ is the **squared Frobenius norm**.

Singular Value Decomposition

Any real matrix $\mathbf{X}_{n\times p}$ can be decomposed as:

$$\mathbf{X} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$$

- $\mathbf{U}_{n\times p}$ is the matrix of orthonormal left singular vectors.
- $\mathbf{\Sigma}_{p\times p}$ is the matrix of positive singular values $\sigma_{11} \geq \sigma_{22} \geq \cdots \geq \sigma_{pp}$.
- $\mathbf{U}_{n\times p}$ is the matrix of orthonormal right singular vectors.

We can reconstruct the original matrix through:

$$\mathbf{X} = \sum_{i}^{p} \sigma_{ii}\mathbf{u}_i\mathbf{v}_i^\top = \sigma_1\mathbf{u}_1\mathbf{v}_1^\top + \sigma_2\mathbf{u}_2\mathbf{v}_2^\top + \cdots + \sigma_p\mathbf{u}_p\mathbf{v}_p^\top$$

Singular Value Decomposition (Theorem)

We can accomplish a rank k approximation of $\mathbf{X}$ in the least squares criterion as:

$$\hat{\mathbf{X}} = \mathbf{U}_{[,1:k]}\mathbf{\Sigma}_{[1:k,1:k]}\mathbf{V}_{[,1:k]}$$

SVD and Spectral decomposition relation

- $\mathbf{X}^\top\mathbf{X} = \mathbf{V}\boldsymbol{\Sigma}\mathbf{U}^\top\mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^\top = \mathbf{V}\mathbf{D}^2\mathbf{V}^\top$
- $\mathbf{X}\mathbf{X}^\top = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^\top\mathbf{V}\boldsymbol{\Sigma}\mathbf{U}^\top = \mathbf{U}\mathbf{D}^2\mathbf{U}^\top$
- The eigenvalues of $\mathbf{X}^\top\mathbf{X}$ and $\mathbf{X}\mathbf{X}^\top$ are squared singular values.
- Singular vectors are eigenvectors: $\mathbf{U}$ of $\mathbf{X}\mathbf{X}^\top$ and $\mathbf{V}$ of $\mathbf{X}^\top\mathbf{X}$
- It turns out that SVD is a generalization of spectral decomposition for non-squared matrices.

Goodness of fit

We need some statistic expressing **goodness-of-fit** (like R^2 in regression). It turns out that the **singular values** are directly informative about the quality of this approximation.

We have that

$$\mathrm{tr}(\mathbf{X}^\top \mathbf{X}) = \mathrm{tr}(\mathbf{V}\boldsymbol{\Sigma}\mathbf{U}^\top \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^\top) = \mathrm{tr}(\mathbf{V}\boldsymbol{\Sigma}^2\mathbf{V}^\top) = \mathrm{tr}(\mathbf{V}^\top\mathbf{V}\boldsymbol{\Sigma}^2) = \mathrm{tr}(\boldsymbol{\Sigma}^2) = \sum_{j=1}^{p} \sigma_{jj}^2 = \sum_{j=1}^{p} \lambda_j$$

And that for a rank 2 approximation:

$$\begin{aligned}
\mathrm{tr}(\widehat{\mathbf{X}}^\top \widehat{\mathbf{X}}) &= \mathrm{tr}(\mathbf{V}_{[,1:2]}\boldsymbol{\Sigma}_{[1:2,1:2]}\mathbf{U}_{[,1:2]}^\top \mathbf{U}_{[,1:2]}\boldsymbol{\Sigma}_{[1:2,1:2]}\mathbf{V}_{[,1:2]}^\top) \\
&= \mathrm{tr}(\mathbf{V}_{[,1:2]}\boldsymbol{\Sigma}_{[1:2,1:2]}^2\mathbf{V}_{[,1:2]}^\top) = \mathrm{tr}(\mathbf{V}_{[,1:2]}^T\mathbf{V}_{[,1:2]}\boldsymbol{\Sigma}_{[1:2,1:2]}^2) \\
&= \mathrm{tr}(\boldsymbol{\Sigma}_{[1:2,1:2]}^2) = \sigma_1^2 + \sigma_2^2 = \lambda_1 + \lambda_2
\end{aligned}$$

Goodness of fit

And that for the error matrix:

$$\mathrm{tr}(\mathbf{E}^\top \mathbf{E}) = \mathrm{tr}((\mathbf{X} - \widehat{\mathbf{X}})^\top (\mathbf{X} - \widehat{\mathbf{X}})) = \mathrm{tr}(\mathbf{V}_{[,3:p]} \mathbf{D}^2_{[3:p,3:p]} \mathbf{V}^\top_{[,3:p]}) = \lambda_3 + \lambda_4 + \cdots + \lambda_p$$

The we can define a goodness-of-fit as:

$$\frac{\mathrm{tr}(\widehat{\mathbf{X}}^\top \widehat{\mathbf{X}})}{\mathrm{tr}(\mathbf{X}^\top \mathbf{X})} = \frac{\lambda_1 + \lambda_2}{\sum_{j=1}^p \lambda_j}$$

Which is similar to the **explained and residual sum of squares** terms in regression.

Weighted singular value decomposition

- We can generalize using **weights** for observations (rows, r_i) and/or variables (columns, c_j)
- We normally minimize $\sum_{i=1}^{n}\sum_{j=1}^{p} e_{ij}^2 = \mathrm{tr}(\mathbf{E}^\top\mathbf{E})$
- Define $\mathbf{D}_r$ with weights for the rows, $\mathbf{D}_c$ with weights for the columns.
- We now wish to minimize $\sum_{i=1}^{n}\sum_{j=1}^{p} r_i c_j e_{ij}^2 = \mathrm{tr}(\mathbf{D}_c\mathbf{E}^\top\mathbf{D}_r\mathbf{E})$
- Note that

$$\sum_{i=1}^{n}\sum_{j=1}^{p} r_i c_j e_{ij}^2 = \sum_{i=1}^{n}\sum_{j=1}^{p}(\sqrt{r_i}\sqrt{c_j}e_{ij})^2 = \sum_{i=1}^{n}\sum_{j=1}^{p}\tilde{e}_{ij}^2$$

$$\sum_{i=1}^{n}\sum_{j=1}^{p}\tilde{e}_{ij}^2 = \sum_{i=1}^{n}\sum_{j=1}^{p}(\sqrt{r_i}\sqrt{c_j}x_{ij} - \sqrt{r_i}\sqrt{c_j}\widehat{x}_{ij})^2$$

Weighted singular value decomposition

- Observe that the solution consists on **transforming** the data prior to the SVD, and **backtransforming** afterwards
$$\mathbf{X}_t = \mathbf{D}_r^{\frac{1}{2}}\mathbf{X}\mathbf{D}_c^{\frac{1}{2}} = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^\top$$
Now compute $\tilde{\mathbf{U}} = \mathbf{D}_r^{-\frac{1}{2}}\mathbf{U}$ and $\tilde{\mathbf{V}} = \mathbf{D}_c^{-\frac{1}{2}}\mathbf{V}$
- Note that $\tilde{\mathbf{U}}\boldsymbol{\Sigma}\tilde{\mathbf{V}}^\top = \mathbf{D}_r^{-\frac{1}{2}}\mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^\top\mathbf{D}_c^{-\frac{1}{2}} = \boldsymbol{\Sigma}_r^{-\frac{1}{2}}\mathbf{D}_r^{\frac{1}{2}}\mathbf{X}\mathbf{D}_c^{\frac{1}{2}}\mathbf{D}_c^{-\frac{1}{2}} = \mathbf{X}$
- $\tilde{\mathbf{U}}_{[,1:k]}\boldsymbol{\Sigma}_{[1:k,1:k]}\tilde{\mathbf{V}}_{[,1:k]}^\top$ is a rank k approximation to $\mathbf{X}$ in the weighted least squares sense.

All MVA methods involved in reducing the number of dimensions of a matrix for interpretability can be formulated through this result. Some articles in the literature uses the term **Generalized Principal Component Analysis**.

Other decompositions

- **Jordan Decomposition**: Block-diagonal form with eigenvalues on the diagonal; uses generalized eigenvectors for defective matrices.
- **Schur Decomposition**: Triangularizes a matrix; diagonal entries are eigenvalues, and Schur vectors generalize eigenvectors.
- **QR Decomposition**: Indirectly used in eigenvalue algorithms (e.g., QR iteration) but does not directly involve eigenvalues.
- **Polar Decomposition**: The positive semi-definite factor has eigenvalues equal to the singular values of the original matrix.
- **Takagi's Factorization**: For complex symmetric matrices; singular values correspond to eigenvalues in modulus (via SVD-like structure).

References

- Johnson and Wichern, (2002) *Applied Multivariate Statistical Analysis*, 6th edition, Prentice Hall, Chapter 2 & 3.
- Jan Graffelman, (2021) *Multivariate Analysis lecture notes, máster interuniversitario en Estadística e Investigación Operativa*.